TECHNICAL WRITING · STANDARD OPERATING PROCEDURE

# Track Inspection & Preventive Maintenance for Urban Light Rail

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## 1. Purpose

This Standard Operating Procedure (SOP) outlines the systematic process for conducting routine track inspections and performing preventive maintenance on urban light rail tracks. The goal is to ensure operational safety, regulatory compliance, and infrastructure longevity.

## 2. Scope

This procedure applies to all TTC-employed maintenance personnel and subcontractors assigned to light rail systems across the network. It includes embedded rail, ballasted track, and special trackwork (e.g., switches, crossings).

## 3. Required Tools & Equipment

- Track geometry measuring tools (gauge, level, straightedge)
- Ultrasonic rail flaw detector
- Rail grinder
- Torque wrench
- PPE (helmet, gloves, safety vest, steel-toe boots, ear protection)
- Flashlight or headlamp for low-visibility inspections
- Digital camera or mobile device for documentation
- Inspection checklists (digital or printed)

## 4. Safety Precautions

- Ensure possession of valid Track Occupancy Permit (TOP) before entering the track area.
- Wear full PPE at all times.
- Maintain radio communication with control center.
- Post warning signage and flaggers where applicable.
- Use spotters when working around active rails or electrified lines.
- Do not proceed with any activity until a proper hazard assessment is conducted.

## 5. Step-by-Step Procedure

5.1 Pre-Inspection Briefing

- Conduct a tailboard meeting to review work scope, hazards, and emergency procedures.
- Confirm that all equipment is calibrated and functional.

5.2 Visual Track Inspection

- Walk the track slowly while scanning for visible defects:
- Cracks, spalling, or wear on railhead
- Loose or missing fasteners
- Misaligned or sunken rail joints
- Ballast condition and drainage issues
- Obstructions, debris, or encroaching vegetation

5.3 Rail Geometry Measurement

- Use a track gauge to measure gauge width and cross-level.
- Document any deviations outside of TTC engineering tolerances.

5.4 Non-Destructive Testing

- Perform ultrasonic testing to detect internal flaws or fatigue cracks.
- Tag any suspected defect areas for follow-up assessment.

5.5 Switch and Crossing Examination

- Test switch throw and locking mechanisms.
- Inspect frog and guardrails for wear and secure bolting.

5.6 Minor Maintenance Tasks

- Tighten loose bolts and clips using a calibrated torque wrench.
- Clear ballast from rail seats and joints.
- Apply lubricant to rail joints as necessary.

5.7 Post-Inspection Review

- Document findings using the TTC Track Inspection Report Template.
- Photograph any significant damage or out-of-spec sections.
- Submit reports to the Infrastructure Control System (ICS) within 24 hours.

## 6. Documentation & Reporting Requirements

- Track Inspection Report (TIR)
- Corrective Action Form (CAF)
- Digital photo logs

● Reports must include date, time, inspector’s name, GPS location (where possible), and specific issue description. 7. Troubleshooting Common Issues

### Issue Possible Cause Immediate Action

### Wide gauge Loose or missing fasteners Tighten bolts, replace as needed

Railhead cracks Fatigue or thermal stress Flag for urgent NDT and reroute traffic

Poor drainage Ballast contamination Schedule ballast cleaning or renewal

### Switch not locking Mechanical obstruction

### Pe Isolate section, escalate to supervisor

## 8. References

- TTC Engineering Standards Manual – Section 3.2

- AREMA Manual for Railway Engineering, Chapter 5
- Ontario Occupational Health and Safety Act (OHSA)
- TTC Safety Handbook for Field Operations